Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 1

KEY WORDS percentage · multiplier · increase · decrease · original amount · discount · GST

A Percentage of an amount ●○○ ANSWER ON THE SHEET

1. Work out:

a) 10% of 350 = ____ **b)** 25% of 84 = ____ **c)** 5% of 260 = ____ **d)** 15% of 80 = ____

e) 40% of $65 = ____ **f)** 12% of 450 = ____ **g)** 7.5% of 240 = ____ **h)** 120% of 35 = ____

B Multipliers ●○○ ANSWER ON THE SHEET

> **Remember:** To increase by 15%, multiply by 1.15. To decrease by 15%, multiply by 0.85.

2. Write the single multiplier for each change.

a) Increase by 20% _____ **b)** Decrease by 20% _____ **c)** Increase by 5% _____

d) Decrease by 35% _____ **e)** Increase by 12.5% _____ **f)** Decrease by 2% _____

g) Increase by 150% _____ **h)** Decrease by 0.5% _____ **i)** Increase by 7.5% _____

3. Describe the change each multiplier makes. For example, × 1.3 is an increase of 30%.

a) × 1.4 _____ **b)** × 0.75 _____ **c)** × 1.08 _____

d) × 0.4 _____ **e)** × 1.015 _____ **f)** × 0.925 _____

C Increasing and decreasing an amount ●●○ WORK IN YOUR BOOK

4. Increase:

a) 60 by 10% **b)** 250 by 30% **c)** $48 by 25%

d) 720 by 5% **e)** $36 by 15% **f)** 1,400 by 2.5%

5. Decrease:

a) 90 by 10% **b)** 400 by 35% **c)** $64 by 25%

d) 850 by 8% **e)** $129 by 40% **f)** 2,600 by 1.5%

D Percentage change ●●○ WORK IN YOUR BOOK

> **Remember:** Percentage change = change ÷ original amount × 100

6. Find the percentage change. Say whether it is an increase or a decrease.

a) from 40 to 50 **b)** from 80 to 68 **c)** from $250 to $290 **d)** from 64 to 48

e) from 1,200 to 1,290 **f)** from $45 to $36 **g)** from 160 to 400 **h)** from 72 to 63

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E Finding the original amount ●●● WORK IN YOUR BOOK

⚠ **Watch out:** The percentage was taken from the original amount, not from the new amount.

7. Find the original amount.

a) After a 10% increase, the amount is 55.

b) After a 20% decrease, the amount is 64.

c) After a 25% increase, the price is $90.

d) After a 30% discount, the price is $84.

e) After a 6% increase, the amount is 1,272.

f) After a 15% decrease, the mass is 34 kg.

g) With 15% GST added, the price is $69.

h) After a 2.5% pay rise, the wage is $28.70.

F Problems ●●● WORK IN YOUR BOOK

8. A jacket costs $180. In a sale it is reduced by 35%. Find the sale price.

9. Mere earns $26 per hour. She gets a 4% pay rise. Find her new hourly rate.

10. A plumber charges $640 plus 15% GST. Find the total bill.

11. A school roll grows from 850 students to 918 students. Find the percentage increase.

12. A phone bought for $1,200 is sold a year later for $780. Find the percentage decrease in its value.

13. A shop buys a jacket for $60 and sells it for $84. Find the percentage profit.

14. A TV costs $1,020 in a sale, after a 15% discount. What was the price before the sale?

15. A café bill is $92, including 15% GST. How much of the bill is GST?

16. Tāne's rent went up by 8% to $486 a week. How much more does he now pay each week?

17. A town has 24,000 people. Its population grows by 5% each year. Find the population after 2 years.

★ Extension FOR EARLY FINISHERS

18. A shop raises a price of $80 by 20%. Later it reduces the new price by 20%.

a) Find the final price.

b) Explain why the final price is not $80.

19. A price rises by 25%. By what percentage must the new price fall to get back to the original price?

20. A car is worth $32,000. It loses 15% of its value each year. Find its value after 3 years.

21. The length of a rectangle is increased by 10% and its width is decreased by 10%. Find the percentage change in its area.

22. Two shops sell the same $400 bike. Shop A takes 25% off. Shop B takes 15% off, then another 10% off the new price. Which shop is cheaper, and by how much?

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Percentage Increase and Decrease

Name _____ Date _____ Calculator allowed Version 2

KEY WORDS percentage · multiplier · increase · decrease · original amount · discount · GST

A Percentage of an amount ●○○ ANSWER ON THE SHEET

1. Work out:

a) 10% of 470 = ____ **b)** 25% of 96 = ____ **c)** 5% of 340 = ____ **d)** 15% of 60 = ____

e) 40% of $95 = ____ **f)** 12% of 350 = ____ **g)** 7.5% of 320 = ____ **h)** 140% of 45 = ____

B Multipliers ●○○ ANSWER ON THE SHEET

> **Remember:** To increase by 15%, multiply by 1.15. To decrease by 15%, multiply by 0.85.

2. Write the single multiplier for each change.

a) Increase by 40% _____ **b)** Decrease by 40% _____ **c)** Increase by 8% _____

d) Decrease by 45% _____ **e)** Increase by 17.5% _____ **f)** Decrease by 4% _____

g) Increase by 120% _____ **h)** Decrease by 1.5% _____ **i)** Increase by 6.5% _____

3. Describe the change each multiplier makes. For example, × 1.3 is an increase of 30%.

a) × 1.6 _____ **b)** × 0.85 _____ **c)** × 1.04 _____

d) × 0.3 _____ **e)** × 1.025 _____ **f)** × 0.875 _____

C Increasing and decreasing an amount ●●○ WORK IN YOUR BOOK

4. Increase:

a) 80 by 10% **b)** 350 by 20% **c)** $56 by 25%

d) 640 by 5% **e)** $44 by 15% **f)** 1,800 by 2.5%

5. Decrease:

a) 70 by 10% **b)** 600 by 45% **c)** $84 by 25%

d) 950 by 6% **e)** $135 by 40% **f)** 3,400 by 1.5%

D Percentage change ●●○ WORK IN YOUR BOOK

> **Remember:** Percentage change = change ÷ original amount × 100

6. Find the percentage change. Say whether it is an increase or a decrease.

a) from 50 to 60 **b)** from 60 to 51 **c)** from $350 to $392 **d)** from 96 to 72

e) from 1,600 to 1,720 **f)** from $65 to $52 **g)** from 140 to 420 **h)** from 56 to 49

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E Finding the original amount ●●● WORK IN YOUR BOOK

⚠ **Watch out:** The percentage was taken from the original amount, not from the new amount.

7. Find the original amount.

a) After a 10% increase, the amount is 77.

b) After a 20% decrease, the amount is 96.

c) After a 25% increase, the price is $60.

d) After a 30% discount, the price is $63.

e) After a 4% increase, the amount is 1,560.

f) After a 15% decrease, the mass is 51 kg.

g) With 15% GST added, the price is $92.

h) After a 2.5% pay rise, the wage is $32.80.

F Problems ●●● WORK IN YOUR BOOK

8. A jacket costs $240. In a sale it is reduced by 35%. Find the sale price.

9. Sione earns $28 per hour. He gets a 3% pay rise. Find his new hourly rate.

10. An electrician charges $520 plus 15% GST. Find the total bill.

11. A school roll grows from 750 students to 795 students. Find the percentage increase.

12. A laptop bought for $1,600 is sold a year later for $1,120. Find the percentage decrease in its value.

13. A shop buys a pair of shoes for $80 and sells them for $108. Find the percentage profit.

14. A fridge costs $1,275 in a sale, after a 15% discount. What was the price before the sale?

15. A restaurant bill is $138, including 15% GST. How much of the bill is GST?

16. Aroha's rent went up by 5% to $504 a week. How much more does she now pay each week?

17. A town has 15,000 people. Its population grows by 6% each year. Find the population after 2 years.

★ Extension FOR EARLY FINISHERS

18. A shop raises a price of $60 by 10%. Later it reduces the new price by 10%.

a) Find the final price.

b) Explain why the final price is not $60.

19. A price falls by 20%. By what percentage must the new price rise to get back to the original price?

20. A car is worth $25,000. It loses 20% of its value each year. Find its value after 3 years.

21. The length of a rectangle is increased by 20% and its width is decreased by 20%. Find the percentage change in its area.

22. Two shops sell the same $600 TV. Shop A takes 30% off. Shop B takes 20% off, then another 10% off the new price. Which shop is cheaper, and by how much?

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